

Murtaza Tahmid Rhythm

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EDUCATION

University of Ottawa

Bachelor of Engineering in Software Engineering, Co-op Program

2nd year — Expected Graduation: Sept 2029

EXPERIENCE

Statistics Canada (Co-op)

Jan 2026 – Present

Software Engineer (Backend)

Ottawa, ON

- Working on modernizing **survey systems** and enabling **hundreds of thousands in annual savings**.
- Leveraged **Python and R** to re-architect survey execution and data-processing logic from SAS for modular, testable, maintainable backend services, improving code reliability and enabling scalable validation workflows.
- Designed and maintained **SQL-backend pipelines** for ingestion, validation, transformation, and persistence at scale.
- Leveraged **Bash and Linux Virtual Machines** to build **automation scripts and backend tooling** to streamline execution, environment setup, and repeatable deployments, which reduced manual configuration overhead.
- Optimized **Python pipelines**, reducing runtime by **30%** and simplifying code structure through vectorized operations.
- Utilized **Git and Gitlab** for version control, and created **documentation** for other developers and client-side users.
- Presented in **cross-functional teams** to review **business logic and program requirements**, implementing modifications and translating rules into deterministic backend logic with validation and reliable outputs.

Industrial Alliance Inc. (Co-op)

May 2025 - Aug 2025

Quality Assurance Technician

Toronto, ON

- Collaborated in a **cross-functional team** to research bottlenecks in the client beneficiary form processing system, analyzing data flows and operational challenges to inform **automation** strategies that improved accuracy and throughput.
- Leveraged **Python’s OCR framework** to develop functions for an **automated parsing filter** that validated client and policy data, reducing bottlenecks and minimizing manual review effort, contributing to **96%** client satisfaction.
- Processed client financial transactions under strict SLAs, maintaining high data integrity through careful validation, while collaborating with client advisors to confirm information accuracy and prevent errors in sensitive financial data.

PROJECTS

PlateCost(AI Receipt Analytics Platform) — CUMSA Hackathon Winner

- Co-developed an end-to-end receipt ingestion and analytics pipeline leveraging **AI** for local restaurant owners.
- Leveraged a **Python OCR framework** to translate receipt images into a **local database** with datasets structured in JSON format, enabling efficient parsing, automated downstream processing, and consistent data availability for analytics.
- Leveraged **TinyLLaMA** to implement LLM-based classification logic in **Python**, which read from a local database, automatically categorized receipt items, generated actionable insights on user expenses, and supported downstream analytics.
- Co-developed a frontend using **React (UMD)** with **HTML, CSS, and JavaScript**, served through **FastAPI static routes**, enabling dynamic, interactive visualizations of user spending patterns and supporting real-time insights for financial tracking.

ConflictZero(Airspace De-confliction Simulator) — UOttawa Hackathon, NAV CANADA Challenge

- Leveraged **Python, Pandas, NumPy**, and **FastAPI** to developed a trajectory simulation engine which processed daily flights.
- Developed high-performance **conflict detection algorithms** identifying horizontal and vertical airplane flight conflicts.
- Exposed **FastAPI REST endpoints** for simulation, conflict analysis, and congestion results enabling frontend integration.
- Built **airspace congestion** and **runway hotspot analytics** with **cargo vs passenger prioritization** logic.
- Implemented **input validation, error handling**, and **deterministic outputs** for synchronization and repeatable analysis.

TECHNICAL SKILLS

Backend & APIs: Python, Java, FastAPI, REST APIs

Data & Analytics: R, SQL, LLMs

Frontend: HTML, React, CSS, JavaScript

DevOps & Systems: Linux, Virtual Machines, Bash Scripting

Tools: Git, GitHub, GitLab, JIRA, Excel, SharePoint

Languages

English: Native

French: Professional

working proficiency